

Supplementary Materials for “Cross-Domain Attribute Alignment with CLIP: A Rehearsal-Free Approach for Class-Incremental Unsupervised Domain Adaptation”

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A Details of Data Construction

In this section, we show the containing classes in each disjoint subset of all three benchmarks (*i.e.*, Office-31 [13], Office-Home [17], and Mini-DomainNet [9]) in Table 1. For Office-31 and Mini-DomainNet, classes are sorted alphabetically; Office-31 is partitioned into groups of 10 categories per step, while Mini-DomainNet is divided into groups of 20 categories per step. For Office-Home, categories are randomly grouped into steps of 10 classes each.

B Details of Training Procedures and Hyperparameters

We use ViT-B/16 [3] as the image encoder for VisTA. The model is trained for 10 epochs per incremental step across all datasets. During training, we employ the mini-batch stochastic gradient descent (SGD) optimizer with an initial learning rate of 3×10^{-3} , adjusting it using a cosine decay schedule. Moreover, the penultimate layer [19] is used for gradient computation in the VAC module, as VisTA employs the ViT-B/16 backbone. We also adopt ViT-B/16 as the image encoder for all baseline methods, including replacing the original backbone (ResNet50 [6]) of ProCA [7] and PLDCA [18]. For AttriCLIP, we learn a shared set of prompts for $\mathcal{D}_s$ and $\mathcal{D}_t$ using Equations (4), (5) in the main paper, and also enhance the quality of pseudo-labeling in $\mathcal{D}_t$ via DeBiasPL.

The γ in Equation (5) of the main paper is set to 0.7 for Office-31 and Mini-DomainNet, and 0.6 for Office-Home. For the loss weights, we set $\lambda_1 = 25, \lambda_2 = 20, \lambda_3 = 1$ for Office-Home and Mini-DomainNet, and $\lambda_1 = 3, \lambda_2 = 6, \lambda_3 = 1$ for Office-31. Additionally, for Office-31 we set $M = 5, N = 13, L = 13$; for Office-Home, $M = 10, N = 8, L = 5$, and for Mini-DomainNet, $M = 10, N = 8, L = 6$.

C More Results of Class-Incremental Unsupervised Domain Adaptation

In this section, we report a comprehensive comparison of VisTA against state-of-the-art baseline methods (*e.g.*, ProCA [7], PLDCA [18], CoOp [20], AttriCLIP [14], AD-CLIP [14], DAMP [4], PGA [10], DAPL [5]), except source-only ViT-B/16 [3] and zero-shot CLIP[12], whose results are

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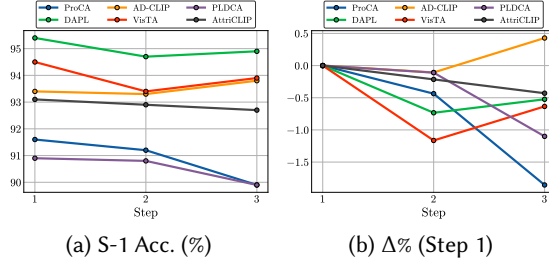


Fig. 1. S-1 Accuracy at each step and its percentage change ($\Delta\%$) compared with Step-1 on Office-31.

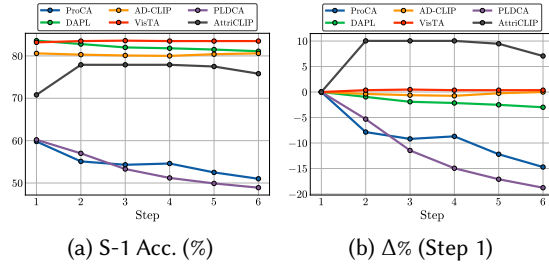


Fig. 2. S-1 Accuracy at each step and its percentage change ($\Delta\%$) compared with Step-1 on Mini-DomainNet.

directly cited from existing paper. Notably, our evaluation adopts more complete and detailed metrics than those in the main paper:

- The Step-level Accuracy on Office-31, Office-Home, and Mini-DomainNet (Tables 4, 5, 6);
- The average S-1 Accuracy of all adaptation tasks at each time step and its percentage change compared with Step-1 on Office-31 and Mini-DomainNet (Figures 1, 2);
- The Step-level Accuracy and S-1 Accuracy for **all adaptation tasks** across Office-31 (Table 7), Office-Home (Tables 8, 9, 10, 11), and Mini-DomainNet (Tables 12, 13, 14, 15).

The experimental results demonstrate that VisTA achieves superior or comparable performance in terms of Step-level Accuracy and S-1 Accuracy across most steps and adaptation tasks. This indicates that VisTA effectively mitigates catastrophic knowledge forgetting while addressing domain shift on Office-31, Office-Home, and Mini-DomainNet.

D Extension to Class-Incremental Source-Free Unsupervised Domain Adaptation

In some practical scenarios, it is infeasible to access source data when training on target domain, while only the pretrained source model are provided [8, 15]. This scenario derives Class-Incremental Source-Free Unsupervised Domain Adaptation (CI-SFUDA) [2]. To address CI-SFUDA, we formalize the training process into two distinct stages: **Source Pretraining** and **Target Deployment**. During Source Pretraining, $\mathcal{A}^s$ is optimized using:

$$\mathcal{L}_{\text{pt}} = \mathcal{L}_{\text{sup}}^s + \lambda_2 \mathcal{L}_{\text{hp}} + \lambda_3 \mathcal{L}_{\text{div}}^s, \quad (1)$$

and $\mathcal{K}^s$ is initialized via K-means clustering before training.

Building on the frozen $[\mathcal{K}^s, \mathcal{A}^s]$, the Target Deployment stage trains $\mathcal{A}^t$ using the following loss where $\mathcal{L}_{\text{con}}$ leverages only $\mathcal{D}^t$:

$$\mathcal{L}_{\text{dp}} = \mathcal{L}_{\text{sup}}^t + \lambda_1 \mathcal{L}_{\text{con}} + \lambda_3 \mathcal{L}_{\text{div}}^t. \quad (2)$$

At each step, VisTA updates $\mathcal{K}^t$ following the protocol in Section 4.1. $\mathcal{L}_{\text{div}}$ is applied separately to $\mathcal{D}_s$ ($\mathcal{L}_{\text{div}}^s$) and $\mathcal{D}_t$ ($\mathcal{L}_{\text{div}}^t$).

In this section, we compare our method with the state-of-the-art CI-SFUDA method GROTO [2], which outperforms the Source-Free version of ProCA [7] and some Source-Free (Universal) Domain Adaptation methods [8, 11, 15, 16]. We directly cite the reported results of GROTO because: (1) VisTA and GROTO share identical data construction protocols on Office-31, Office-Home, and Mini-DomainNet benchmarks, and (2) GROTO employs the same ViT-B/16 backbone as VisTA. Following the evaluation protocol of CI-UDA, we measure performance via Final Accuracy (Tables 16, 17, 18) across three benchmarks. For Office-31 and Office-Home, we additionally report Step-level Accuracy (Table 19) and S-1 Accuracy (Figures 3, 4), since GROTO exclusively reports averaged Final Accuracy across three subtasks per target domain on Mini-DomainNet.

The results demonstrate that the source-free version of VisTA achieves improvements of 2.2% and 19.1% over GROTO on Office-Home and Mini-DomainNet, respectively. Notably, it surpasses GROTO in the last S-1 Accuracy on Office-31 and Office-Home, with significantly smaller degradation in S-1 Accuracy than that of GROTO. These findings validate that VisTA—despite not being explicitly designed for CI-SFUDA—either outperforms or achieves performance comparable to GROTO across most metrics. Furthermore, these results reinforce our argument in Section 5.2 that the performance advantages of VisTA scale with benchmark complexity. The findings also confirm the capability of VisTA to mitigate domain shift and catastrophic forgetting when extended to source-free scenarios.

E More Ablation Analysis

We compare the computational overhead (including GPU Memory Usage and Frames Per Second at batch size 1) in deployment between VisTA and some baseline methods. Table 2 summarizes the results. Compared with CLIP-based methods (e.g., CoOp, AttriCLIP, and AD-CLIP), VisTA maintains comparable computational efficiency while achieving significantly higher accuracy. In contrast to the CI-UDA approach PLDCA, VisTA incurs increased time due to the use of CLIP, but substantially outperforms PLDCA in performance.

To analyze the performance of methods across different Vision-Language Models (weaker or stronger), we evaluate VisTA against some CLIP-based baseline methods on weaker models: CLIP ViT-B/32; default models: CLIP ViT-B/16; and stronger models: OpenCLIP [1] ViT-B/16 (LAION400M) and CLIP ViT-L/14. Table 3 demonstrates that the performance of all methods depends on the selection of Vision-Language Models. However, VisTA consistently achieves the best performance. For the weaker model (CLIP ViT-B/32), though the performance is inferior to the default or stronger model, VisTA still obtains good accuracy and outperforms all baseline methods remarkably.

Table 1. Class names in each time step on Office-31, Office-Home, and Mini-DomainNet.

Dataset	Time Step	Class Index	Class Name
Office-31 [13]	Step 1	[0 ~ 9]	back pack, bike, bike helmet, bookcase, bottle, calculator, desk chair, desk lamp, desktop computer, file cabinet
	Step 2	[10 ~ 19]	headphones, keyboard, laptop computer, letter tray, mobile phone, monitor, mouse, mug, paper notebook, pen
	Step 3	[20 ~ 29]	phone, printer, projector, punchers, ring binder, ruler, scissors, speaker, stapler, tape dispenser
Office-Home [17]	Step 1	[0 ~ 9]	Drill, Exit Sign, Bottle, Glasses, Computer, File Cabinet, Shelf, Toys, Sink, Laptop
	Step 2	[10 ~ 19]	Kettle, Folder, Keyboard, Flipflops, Pencil, Bed, Hammer, ToothBrush, Couch, Bike
	Step 3	[20 ~ 29]	Postit Notes, Mug, Webcam, Desk Lamp, Telephone, Helmet, Mouse, Pen, Monitor, Mop
	Step 4	[30 ~ 39]	Sneakers, Notebook, Backpack, Alarm Clock, Push Pin, Paper Clip, Batteries, Radio, Fan, Ruler
	Step 5	[40 ~ 49]	Pan, Screwdriver, Trash Can, Printer, Speaker, Eraser, Bucket, Chair, Calendar, Calculator
	Step 6	[50 ~ 59]	Flowers, Lamp Shade, Spoon, Candles, Clipboards Scissors, TV, Curtains, Fork, Soda
Mini-DomainNet [9]	Step 1	[0 ~ 19]	aircraft carrier, alarm clock, ant, anvil, asparagus, axe, banana, basket, bathtub, bear, bee, bird, blackberry, blueberry, bottlecap, broccoli, bus, butterfly, cactus, cake
	Step 2	[20 ~ 39]	calculator, camel, camera, candle, cannon, canoe, carrot, castle, cat, ceiling fan, cello, cell phone, chair, chandelier, coffee cup, compass, computer, cow, crab, crocodile
	Step 3	[40 ~ 59]	cruise_ship, dog, dolphin, dragon, drums, duck, dumbbell, elephant, eyeglasses, feather, fence, fish, flamingo, flower, foot, fork, frog, giraffe, goatee, grapes
	Step 4	[60 ~ 79]	guitar, hammer, helicopter, helmet, horse, kangaroo, lantern, laptop, leaf, lion, lipstick, lobster, microphone, monkey, mosquito, mouse, mug, mushroom, onion, panda
	Step 5	[80 ~ 99]	peanut, pear, peas, pencil, penguin, pig, pillow, pineapple, potato, power outlet, purse, rabbit, raccoon, rhinoceros, rifle, saxophone, screwdriver, sea turtle, see saw, sheep
	Step 6	[100 ~ 119]	shoe, skateboard, snake, speedboat, spider, squirrel, strawberry, streetlight, string bean, submarine, swan, table, teapot, teddy-bear, television, The Eiffel Tower, The Great Wall of China, tiger, toe, train

Table 2. Computational overhead on Office-Home.

Method	GPU Memory Usage (MB) ↓	Frames Per Second ↑	Final Accuracy (%)
PLDCA (TIP24)	1435.9	144.8	76.6
CoOp (IJCv22)	440.7	47.5	82.7
AttriCLIP (CVPR23)	705.3	48.1	82.3
AD-CLIP (ICCVW23)	1016.2	11.2	83.7
VisTA (Ours)	885.6	43.9	85.3

Table 3. Performance across different Vision-Language Models.

Method	CLIP B/32	CLIP B/16	OpenCLIP	CLIP L/14
vanilla (ICML21)	79.1	82.4	85.5	87.0
CoOp (IJCV22)	79.3	82.7	83.8	87.6
AttriCLIP (CVPR23)	78.8	82.3	83.1	86.8
AD-CLIP (ICCVW23)	80.1	83.7	82.0	85.2
VisTA (Ours)	82.2	85.3	86.8	89.5

Table 4. Step-level Accuracy (%) on Office-31. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Method	DA	CI	RF	Step 1	Step 2	Step 3	Avg.
ProCA (ECCV22)	✓	✓	✗	92.9	90.9	88.0	90.6
PLDCA (TIP24)	✓	✓	✗	91.5	90.9	88.7	90.3
CoOp (IJCV22)	–	–	–	93.5	93.0	87.9	91.5
AttriCLIP (CVPR23)	✗	✓	✓	93.1	91.1	86.3	90.1
AD-CLIP (ICCVW23)	✓	✗	✓	93.3	89.2	87.0	89.8
DAMP (CVPR24)	✓	✗	✓	93.7	89.9	84.6	89.4
PGA (NeurIPS24)	✓	✗	✓	92.2	79.4	75.1	82.2
DAPL (TNNLS25)	✓	✗	✓	95.4	86.6	79.2	87.0
VisTA (Ours)	✓	✓	✓	94.3	93.5	89.3	92.4

Table 5. Step-level Accuracy (%) on Office-Home. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
ProCA (ECCV22)	✓	✓	✗	73.2	74.0	72.3	73.3	73.8	74.1	73.5
PLDCA (TIP24)	✓	✓	✗	70.8	74.4	73.7	74.8	76.1	76.6	74.4
CoOp (IJCV22)	–	–	–	79.2	82.1	81.6	82.7	82.8	82.7	81.9
AttriCLIP (CVPR23)	✗	✓	✓	82.7	84.5	81.8	82.7	83.1	82.3	82.9
AD-CLIP (ICCVW23)	✓	✗	✓	79.5	83.5	81.8	83.5	83.3	83.8	82.6
DAMP (CVPR23)	✓	✗	✓	81.0	83.3	82.1	82.4	82.0	81.9	82.1
PGA (NeurIPS24)	✓	✗	✓	78.7	73.9	71.1	71.4	71.0	70.5	72.8
DAPL (TNNLS25)	✓	✗	✓	81.6	83.7	81.9	82.9	83.4	83.4	82.8
VisTA (Ours)	✓	✓	✓	80.3	84.7	84.2	85.1	85.1	85.3	84.1

Table 6. Step-level Accuracy (%) on Mini-DomainNet. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
ProCA (ECCV22)	✓	✓	✗	59.8	55.3	57.8	57.1	55.1	56.3	56.9
PLDCA (TIP24)	✓	✓	✗	61.0	60.9	63.7	61.4	60.4	61.2	61.4
CoOp (IJCV22)	–	–	–	80.3	81.2	83.2	82.7	81.7	81.2	81.7
AttriCLIP (CVPR23)	✗	✓	✓	74.1	77.9	80.2	79.9	80.0	78.0	78.4
AD-CLIP (ICCVW23)	✓	✗	✓	81.5	83.2	84.4	83.9	83.6	82.1	83.1
DAMP (CVPR24)	✓	✗	✓	82.5	83.5	84.2	83.2	82.3	80.2	82.6
PGA (NeurIPS24)	✓	✗	✓	16.0	30.3	48.7	60.8	77.9	80.9	52.4
DAPL (TNNLS25)	✓	✗	✓	83.7	85.6	86.3	85.5	84.3	83.7	84.9
VisTA (Ours)	✓	✓	✓	83.5	85.5	87.2	86.6	86.1	85.4	85.7

Table 7. Classification accuracies (%) on Office-31. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Avg.
A→D	ProCA (ECCV22)	✓	✓	✗	97.4 (97.2)	90.6 (97.2)	91.5 (96.8)	93.2 (97.1)
	PLDCA (TIP24)	✓	✓	✗	90.9 (91.4)	90.0 (91.4)	91.5 (91.4)	90.8 (91.4)
	CoOp (IJCV22)	–	–	–	97.4 (96.8)	93.5 (97.4)	89.0 (97.4)	93.3 (97.4)
	AttriCLIP (CVPR23)	✗	✓	✓	96.1 (96.1)	71.5 (93.0)	81.8 (89.1)	83.1 (92.7)
	AD-CLIP (ICCVW23)	✓	✗	✓	95.5 (96.1)	90.3 (96.8)	82.8 (96.1)	89.2 (96.3)
	DAMP (CVPR24)	✓	✗	✓	96.1 (95.5)	89.4 (95.5)	85.1 (95.5)	90.2 (95.5)
	PGA (NeurIPS24)	✓	✗	✓	93.0 (93.0)	62.9 (67.2)	49.0 (79.7)	68.3 (80.0)
	DAPL (TNNLS25)	✓	✗	✓	96.8 (96.8)	87.7 (96.8)	79.1 (97.4)	87.9 (97.0)
	VisTA (Ours)	✓	✓	✓	98.7 (98.7)	92.3 (96.8)	87.8 (97.4)	92.9 (97.6)
A→W	ProCA (ECCV22)	✓	✓	✗	94.5 (95.8)	87.7 (95.8)	86.3 (95.3)	89.5 (95.6)
	PLDCA (TIP24)	✓	✓	✗	94.0 (95.5)	90.0 (95.8)	90.7 (95.8)	91.6 (95.7)
	CoOp (IJCV22)	–	–	–	96.6 (96.6)	92.6 (96.6)	87.0 (96.6)	92.1 (96.6)
	AttriCLIP (CVPR23)	✗	✓	✓	98.7 (98.7)	92.6 (99.1)	85.7 (97.4)	92.3 (98.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	99.1 (99.1)	82.0 (93.6)	85.9 (99.6)	89.0 (97.4)
	DAMP (CVPR24)	✓	✗	✓	99.1 (99.1)	85.4 (96.6)	79.3 (94.0)	87.9 (96.6)
	PGA (NeurIPS24)	✓	✗	✓	96.9 (96.9)	67.8 (49.2)	68.4 (75.0)	77.7 (73.7)
	DAPL (TNNLS25)	✓	✗	✓	98.3 (98.3)	84.7 (97.0)	76.7 (97.9)	86.6 (97.7)
	VisTA (Ours)	✓	✓	✓	100.0 (100.0)	94.9 (98.3)	88.2 (98.3)	94.4 (98.9)
D→A	ProCA (ECCV22)	✓	✓	✗	81.5 (77.3)	84.5 (76.0)	75.6 (71.8)	80.6 (75.0)
	PLDCA (TIP24)	✓	✓	✗	81.8 (77.5)	84.0 (78.2)	75.6 (74.9)	80.5 (76.9)
	CoOp (IJCV22)	–	–	–	85.1 (85.1)	90.4 (85.1)	83.0 (85.1)	86.2 (86.2)
	AttriCLIP (CVPR23)	✗	✓	✓	84.0 (84.0)	81.3 (84.0)	79.5 (83.4)	81.6 (83.8)
	AD-CLIP (ICCVW23)	✓	✗	✓	85.0 (85.0)	84.1 (89.5)	83.0 (87.1)	84.0 (87.2)
	DAMP (CVPR24)	✓	✗	✓	85.3 (85.3)	89.5 (84.5)	81.3 (87.1)	85.4 (85.6)
	PGA (NeurIPS24)	✓	✗	✓	86.5 (86.5)	88.3 (86.1)	81.2 (84.1)	85.3 (85.6)
	DAPL (TNNLS25)	✓	✗	✓	90.9 (90.9)	88.3 (89.6)	79.5 (90.8)	86.2 (90.4)
	VisTA (Ours)	✓	✓	✓	85.6 (85.6)	90.9 (86.3)	84.5 (86.9)	87.0 (86.3)
D→W	ProCA (ECCV22)	✓	✓	✗	100.0 (100.0)	99.8 (100.0)	98.3 (100.0)	99.4 (100.0)
	PLDCA (TIP24)	✓	✓	✗	100.0 (100.0)	97.3 (100.0)	96.9 (100.0)	98.1 (100.0)
	CoOp (IJCV22)	–	–	–	98.7 (98.7)	98.3 (98.7)	96.1 (98.7)	97.7 (97.9)
	AttriCLIP (CVPR23)	✗	✓	✓	98.7 (98.7)	98.1 (98.7)	94.4 (99.6)	97.1 (99.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	97.0 (97.0)	97.0 (98.7)	95.1 (99.1)	96.4 (98.3)
	DAMP (CVPR24)	✓	✗	✓	99.6 (99.6)	90.0 (99.6)	90.3 (99.1)	93.3 (99.4)
	PGA (NeurIPS24)	✓	✗	✓	93.8 (93.8)	96.5 (100)	88.7 (99.2)	93.0 (97.7)
	DAPL (TNNLS25)	✓	✗	✓	99.1 (99.1)	83.0 (98.3)	79.2 (97.4)	87.1 (98.3)
	VisTA (Ours)	✓	✓	✓	100.0 (100.0)	97.7 (98.7)	95.2 (99.6)	97.6 (99.4)
W→A	ProCA (ECCV22)	✓	✓	✗	83.7 (79.2)	83.2 (77.9)	77.2 (75.6)	81.4 (77.6)
	PLDCA (TIP24)	✓	✓	✗	82.0 (81.2)	84.9 (79.4)	77.8 (77.5)	81.6 (79.4)
	CoOp (IJCV22)	–	–	–	84.7 (84.7)	87.8 (84.7)	81.5 (84.7)	84.7 (84.7)
	AttriCLIP (CVPR23)	✗	✓	✓	82.9 (82.9)	87.0 (84.2)	79.8 (84.0)	83.2 (83.7)
	AD-CLIP (ICCVW23)	✓	✗	✓	85.6 (85.6)	88.4 (84.5)	81.4 (84.0)	85.1 (84.7)
	DAMP (CVPR24)	✓	✗	✓	86.7 (86.7)	90.6 (86.2)	80.1 (88.2)	85.8 (87.0)
	PGA (NeurIPS24)	✓	✗	✓	86.6 (86.6)	89.2 (85.8)	81.3 (85.2)	85.7 (85.9)
	DAPL (TNNLS25)	✓	✗	✓	90.4 (90.4)	88.8 (89.4)	81.1 (89.9)	86.8 (89.9)
	VisTA (Ours)	✓	✓	✓	85.2 (85.2)	90.5 (85.8)	84.0 (85.9)	86.6 (85.6)
W→D	ProCA (ECCV22)	✓	✓	✗	100.0 (100.0)	99.7 (100.0)	99.4 (100.0)	99.8 (100.0)
	PLDCA (TIP24)	✓	✓	✗	100.0 (100.0)	99.4 (100.0)	99.4 (100.0)	99.6 (100.0)
	CoOp (IJCV22)	–	–	–	98.1 (98.1)	97.7 (98.1)	98.1 (98.1)	98.0 (98.1)
	AttriCLIP (CVPR23)	✗	✓	✓	97.4 (97.4)	96.8 (96.1)	95.4 (96.8)	96.5 (96.8)
	AD-CLIP (ICCVW23)	✓	✗	✓	97.4 (97.4)	94.5 (96.8)	93.6 (96.8)	95.2 (97.0)
	DAMP (CVPR24)	✓	✗	✓	95.5 (95.5)	94.5 (92.9)	91.7 (96.1)	93.9 (94.8)
	PGA (NeurIPS24)	✓	✗	✓	96.1 (96.1)	71.5 (93.0)	81.8 (89.1)	83.1 (92.7)
	DAPL (TNNLS25)	✓	✗	✓	96.8 (96.8)	87.1 (96.8)	79.3 (96.1)	87.7 (96.6)
	VisTA (Ours)	✓	✓	✓	97.4 (97.4)	95.2 (94.2)	96.5 (95.5)	96.4 (95.7)

Table 8. Classification accuracies (%) on Office-Home with Art as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
A→C	ProCA (ECCV22)	✓	✓	✗	58.9 (59.4)	60.3 (57.5)	55.7 (56.9)	56.9 (58.5)	56.8 (55.8)	56.8 (55.3)	57.6 (57.2)
	PLDCA (TIP24)	✓	✓	✗	57.8 (62.3)	58.1 (57.8)	54.7 (55.6)	58.1 (57.0)	58.7 (55.2)	58.3 (53.4)	57.6 (56.9)
	CoOp (IJCV22)	-	-	-	65.4 (65.4)	71.1 (65.4)	67.4 (65.4)	69.4 (65.4)	70.7 (65.4)	70.6 (65.4)	69.1 (65.4)
	AttriCLIP (CVPR23)	✗	✓	✓	63.1 (63.1)	71.1 (60.2)	70.1 (61.4)	68.3 (59.3)	68.0 (60.1)	67.4 (56.1)	68.0 (60.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	68.5 (68.5)	71.2 (62.8)	68.4 (62.0)	71.1 (63.1)	70.0 (60.1)	70.9 (61.1)	70.0 (62.9)
	DAMP (CVPR24)	✓	✗	✓	67.5 (67.5)	70.6 (63.5)	68.4 (67.0)	69.7 (66.1)	68.2 (61.3)	68.3 (64.3)	68.8 (65.0)
	PGA (NeurIPS24)	✓	✗	✓	55.8 (55.8)	64.6 (50.5)	58.5 (51.6)	58.4 (51.9)	61.8 (52.3)	61.6 (45.0)	60.1 (51.2)
	DAPL (TNNLS25)	✓	✗	✓	61.0 (61.0)	72.0 (62.6)	67.7 (61.9)	70.0 (62.9)	70.4 (59.3)	69.3 (60.5)	68.4 (61.4)
	VisTA (Ours)	✓	✓	✓	61.9 (61.9)	72.2 (63.5)	70.4 (64.4)	71.9 (64.1)	72.4 (63.4)	71.8 (64.8)	70.1 (63.7)
A→P	ProCA (ECCV22)	✓	✓	✗	89.1 (86.7)	84.5 (85.2)	80.3 (84.6)	81.9 (83.2)	82.1 (82.5)	81.9 (83.9)	83.3 (84.4)
	PLDCA (TIP24)	✓	✓	✗	79.7 (79.1)	82.0 (77.2)	80.6 (75.6)	83.9 (76.6)	83.5 (75.5)	83.2 (73.8)	82.1 (76.3)
	CoOp (IJCV22)	-	-	-	86.1 (86.1)	87.4 (86.1)	88.2 (86.1)	89.1 (86.1)	88.9 (86.1)	88.9 (86.1)	88.1 (86.1)
	AttriCLIP (CVPR23)	✗	✓	✓	92.1 (92.1)	90.5 (84.8)	86.6 (88.6)	89.4 (88.3)	89.8 (85.4)	87.9 (86.7)	89.4 (87.7)
	AD-CLIP (ICCVW23)	✓	✗	✓	90.1 (90.1)	88.2 (87.1)	86.4 (80.3)	91.3 (86.5)	90.8 (88.8)	91.2 (88.6)	89.7 (86.9)
	DAMP (CVPR24)	✓	✗	✓	92.1 (92.1)	86.9 (87.9)	89.1 (86.4)	89.1 (85.1)	89.1 (83.6)	86.7 (80.3)	88.8 (85.9)
	PGA (NeurIPS24)	✓	✗	✓	87.8 (87.8)	79.0 (72.3)	84.2 (73.9)	77.4 (58.3)	80.2 (75.0)	70.9 (66.1)	79.9 (72.2)
	DAPL (TNNLS25)	✓	✗	✓	88.5 (88.5)	85.4 (83.3)	89.5 (87.7)	90.8 (89.1)	91.5 (87.1)	91.2 (88.8)	89.5 (87.4)
	VisTA (Ours)	✓	✓	✓	91.8 (91.8)	91.7 (91.5)	89.1 (91.8)	92.9 (91.8)	93.1 (91.5)	92.8 (92.4)	92.4 (91.8)
A→R	ProCA (ECCV22)	✓	✓	✗	86.5 (86.7)	90.2 (86.1)	88.2 (85.1)	89.2 (84.8)	90.0 (83.6)	89.9 (83.8)	89.0 (85.0)
	PLDCA (TIP24)	✓	✓	✗	86.1 (85.8)	90.7 (84.4)	89.0 (82.5)	88.9 (82.1)	88.8 (82.2)	89.5 (81.5)	88.8 (83.1)
	CoOp (IJCV22)	-	-	-	86.2 (86.2)	89.2 (86.2)	89.0 (86.2)	89.8 (86.2)	89.9 (86.2)	89.8 (86.2)	89.0 (86.2)
	AttriCLIP (CVPR23)	✗	✓	✓	88.3 (88.3)	90.2 (85.2)	90.0 (84.3)	90.3 (87.1)	90.8 (86.4)	89.4 (85.8)	89.8 (86.2)
	AD-CLIP (ICCVW23)	✓	✗	✓	84.8 (84.8)	89.4 (85.4)	85.6 (83.3)	89.7 (84.2)	89.4 (84.8)	89.3 (85.7)	88.0 (84.7)
	DAMP (CVPR24)	✓	✗	✓	87.7 (87.7)	89.4 (85.5)	88.5 (85.1)	89.7 (86.5)	88.5 (83.3)	88.7 (84.2)	88.8 (85.4)
	PGA (NeurIPS24)	✓	✗	✓	89.2 (89.2)	81.0 (71.4)	82.5 (70.9)	78.7 (79.8)	83.3 (71.4)	85.3 (74.4)	83.3 (76.2)
	DAPL (TNNLS25)	✓	✗	✓	90.7 (90.7)	90.3 (86.7)	89.3 (89.1)	90.0 (87.1)	90.3 (85.8)	90.5 (86.4)	90.2 (87.6)
	VisTA (Ours)	✓	✓	✓	86.4 (86.4)	90.3 (86.8)	90.5 (86.8)	90.9 (86.5)	91.1 (86.4)	91.3 (87.0)	90.1 (86.6)

Table 9. Classification accuracies (%) on Office-Home with Clipart as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
C→A	ProCA (ECCV22)	✓	✓	✗	67.6 (67.2)	66.9 (66.2)	64.7 (67.7)	65.9 (67.4)	66.3 (66.2)	68.4 (66.7)	66.6 (66.9)
	PLDCA (TIP24)	✓	✓	✗	67.1 (70.6)	71.5 (66.8)	70.7 (66.6)	70.1 (64.0)	73.8 (66.9)	76.5 (64.6)	71.6 (66.6)
	CoOp (IJCV22)	-	-	-	81.3 (81.3)	82.0 (81.3)	80.5 (81.3)	80.9 (81.3)	80.6 (81.3)	81.4 (81.3)	81.1 (81.3)
	AttriCLIP (CVPR23)	✗	✓	✓	83.5 (83.5)	83.8 (80.0)	81.1 (78.5)	82.7 (81.0)	81.8 (79.7)	84.0 (79.7)	82.8 (80.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	78.0 (78.0)	84.5 (82.3)	79.0 (72.2)	82.1 (78.2)	81.5 (78.0)	82.2 (81.0)	81.2 (78.3)
	DAMP (CVPR24)	✓	✗	✓	73.9 (73.9)	83.9 (80.0)	81.3 (77.8)	81.9 (78.5)	78.6 (70.4)	81.3 (76.7)	80.2 (76.2)
	PGA (NeurIPS24)	✓	✗	✓	84.9 (84.9)	50.8 (13.0)	73.8 (64.1)	62.6 (72.4)	60.0 (33.1)	55.7 (51.0)	64.6 (53.1)
	DAPL (TNNLS25)	✓	✗	✓	83.0 (83.0)	85.8 (80.3)	82.7 (80.3)	82.7 (78.2)	82.8 (78.7)	83.3 (78.2)	83.4 (79.8)
	VisTA (Ours)	✓	✓	✓	78.2 (78.2)	84.9 (78.7)	83.3 (78.7)	83.8 (79.5)	83.1 (79.2)	84.4 (80.8)	83.0 (79.2)
C→P	ProCA (ECCV22)	✓	✓	✗	82.5 (83.5)	80.0 (82.5)	79.5 (82.7)	80.5 (82.1)	81.1 (82.0)	80.9 (81.9)	80.8 (82.4)
	PLDCA (TIP24)	✓	✓	✗	80.0 (82.2)	79.4 (81.6)	80.2 (82.0)	81.2 (81.5)	81.6 (80.7)	83.3 (80.4)	81.0 (81.4)
	CoOp (IJCV22)	-	-	-	90.6 (90.6)	87.6 (90.6)	88.7 (90.6)	90.0 (90.6)	89.9 (90.6)	88.4 (90.6)	89.2 (90.6)
	AttriCLIP (CVPR23)	✗	✓	✓	92.8 (92.8)	93.8 (91.8)	89.3 (87.1)	90.1 (89.4)	89.2 (88.6)	90.2 (89.4)	90.9 (89.9)
	AD-CLIP (ICCVW23)	✓	✗	✓	92.2 (92.2)	89.8 (90.4)	92.3 (90.9)	91.5 (89.4)	91.6 (91.8)	91.3 (91.0)	91.5 (91.0)
	DAMP (CVPR24)	✓	✗	✓	93.1 (93.1)	88.5 (89.1)	90.4 (90.7)	91.3 (89.5)	90.8 (87.1)	90.1 (89.4)	90.7 (89.8)
	PGA (NeurIPS24)	✓	✗	✓	87.3 (87.3)	83.3 (81.7)	82.3 (63.8)	83.1 (80.3)	85.7 (73.1)	83.7 (83.8)	84.2 (78.3)
	DAPL (TNNLS25)	✓	✗	✓	89.4 (89.4)	85.9 (87.3)	89.4 (88.5)	90.5 (90.1)	91.1 (88.0)	90.1 (84.6)	89.4 (88.0)
	VisTA (Ours)	✓	✓	✓	92.1 (92.1)	91.5 (92.1)	92.3 (92.2)	93.6 (92.7)	93.4 (92.7)	92.9 (92.8)	92.6 (92.4)
C→R	ProCA (ECCV22)	✓	✓	✗	79.5 (79.7)	84.3 (77.5)	84.6 (78.3)	83.1 (75.8)	83.7 (76.2)	83.9 (77.2)	83.2 (77.5)
	PLDCA (TIP24)	✓	✓	✗	73.7 (74.8)	80.4 (73.0)	80.6 (70.8)	81.5 (70.6)	84.1 (71.5)	85.3 (70.9)	80.9 (71.9)
	CoOp (IJCV22)	-	-	-	84.3 (84.3)	87.6 (84.3)	86.9 (84.3)	87.9 (84.3)	88.1 (84.3)	87.9 (84.3)	87.1 (84.3)
	AttriCLIP (CVPR23)	✗	✓	✓	88.0 (88.0)	90.9 (86.7)	90.1 (87.0)	89.8 (87.1)	89.0 (82.4)	87.3 (74.7)	89.2 (84.3)
	AD-CLIP (ICCVW23)	✓	✗	✓	87.6 (87.6)	85.1 (78.9)	88.1 (87.1)	89.7 (87.6)	87.8 (81.3)	90.4 (85.9)	88.1 (84.7)
	DAMP (CVPR24)	✓	✗	✓	88.3 (88.3)	89.7 (86.4)	89.0 (86.5)	88.6 (86.4)	87.3 (85.7)	88.2 (84.8)	88.5 (86.4)
	PGA (NeurIPS24)	✓	✗	✓	88.8 (88.8)	81.2 (68.1)	82.3 (75.5)	82.0 (72.0)	79.6 (64.5)	84.3 (74.1)	83.0 (73.8)
	DAPL (TNNLS25)	✓	✗	✓	90.9 (90.9)	91.1 (88.2)	89.4 (88.8)	89.0 (87.4)	88.8 (80.3)	90.1 (86.2)	89.9 (87.0)
	VisTA (Ours)	✓	✓	✓	87.0 (87.0)	89.9 (87.1)	90.3 (87.4)	90.8 (87.6)	90.8 (87.9)	91.1 (87.7)	90.0 (87.5)

Table 10. Classification accuracies (%) on Office-Home with Product as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
P→A	ProCA (ECCV22)	✓	✓	✗	63.8 (66.0)	68.1 (65.9)	66.4 (65.4)	68.7 (65.7)	68.8 (66.2)	70.1 (65.0)	67.6 (65.7)
	PLDCA (TIP24)	✓	✓	✗	63.3 (64.1)	68.6 (62.7)	67.8 (62.3)	68.6 (63.3)	72.6 (63.5)	74.2 (62.6)	69.2 (63.1)
	CoOp (IJCV22)	–	–	–	72.2 (72.2)	78.3 (72.2)	78.4 (72.2)	79.1 (72.2)	78.8 (72.2)	79.8 (72.2)	77.8 (72.2)
	AttriCLIP (CVPR23)	✗	✓	✓	82.5 (82.5)	85.4 (81.5)	82.8 (79.2)	82.2 (77.5)	80.8 (78.7)	82.3 (74.7)	82.7 (79.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	73.7 (73.7)	80.9 (75.4)	75.2 (78.9)	79.6 (72.9)	80.8 (76.7)	80.4 (77.2)	78.4 (75.8)
	DAMP (CVPR24)	✓	✗	✓	76.5 (76.5)	80.2 (71.9)	80.0 (77.0)	79.9 (71.1)	79.1 (72.2)	81.5 (77.5)	79.5 (74.4)
	PGA (NeurIPS24)	✓	✗	✓	83.3 (83.3)	70.8 (54.9)	64.8 (56.0)	54.9 (38.5)	67.2 (62.0)	58.5 (53.9)	66.6 (58.1)
	DAPL (TNNLS25)	✓	✗	✓	83.8 (83.8)	84.5 (78.5)	81.8 (79.2)	82.8 (82.5)	83.3 (79.5)	83.3 (78.7)	83.3 (80.4)
	VisTA (Ours)	✓	✓	✓	81.0 (81.0)	84.7 (79.5)	83.9 (80.0)	84.5 (80.0)	83.1 (79.5)	84.7 (80.3)	83.7 (80.0)
P→C	ProCA (ECCV22)	✓	✓	✗	61.3 (59.5)	52.1 (59.7)	48.4 (57.8)	48.9 (57.4)	51.5 (58.1)	51.3 (56.7)	52.3 (58.2)
	PLDCA (TIP24)	✓	✓	✗	56.3 (54.5)	50.3 (51.7)	50.5 (48.9)	50.8 (46.9)	54.2 (46.6)	54.5 (45.0)	52.8 (48.9)
	CoOp (IJCV22)	–	–	–	66.3 (66.3)	69.4 (66.3)	66.7 (66.3)	68.8 (66.3)	69.6 (66.3)	69.6 (66.3)	68.4 (66.3)
	AttriCLIP (CVPR23)	✗	✓	✓	70.3 (70.3)	74.2 (70.7)	71.9 (66.1)	69.8 (61.6)	70.3 (62.6)	67.3 (55.2)	70.6 (64.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	71.9 (71.9)	73.0 (68.8)	70.7 (69.9)	71.0 (67.6)	72.0 (69.3)	71.9 (70.2)	71.8 (69.6)
	DAMP (CVPR24)	✓	✗	✓	66.6 (66.6)	76.4 (72.0)	69.9 (66.1)	68.5 (66.9)	70.2 (64.4)	68.3 (67.0)	70.0 (67.2)
	PGA (NeurIPS24)	✓	✗	✓	56.4 (56.4)	67.3 (58.6)	62.4 (52.0)	55.4 (42.0)	63.6 (45.9)	59.7 (46.3)	60.8 (50.2)
	DAPL (TNNLS25)	✓	✗	✓	63.5 (63.5)	70.7 (61.3)	67.0 (59.0)	68.4 (62.2)	69.5 (60.5)	68.9 (59.2)	68.0 (60.9)
	VisTA (Ours)	✓	✓	✓	65.1 (65.1)	72.2 (64.1)	70.7 (65.7)	72.6 (65.8)	73.3 (66.1)	72.8 (66.4)	71.1 (65.5)
P→R	ProCA (ECCV22)	✓	✓	✗	85.2 (85.3)	89.5 (83.4)	86.5 (83.1)	88.7 (83.2)	89.1 (81.6)	89.9 (79.9)	88.1 (82.8)
	PLDCA (TIP24)	✓	✓	✗	84.8 (84.9)	87.8 (82.9)	89.0 (80.5)	89.1 (80.4)	89.2 (78.6)	87.6 (77.9)	87.9 (80.9)
	CoOp (IJCV22)	–	–	–	85.1 (85.1)	88.2 (85.1)	87.8 (85.1)	88.9 (85.1)	89.2 (85.1)	89.6 (85.1)	88.1 (85.1)
	AttriCLIP (CVPR23)	✗	✓	✓	88.3 (88.3)	88.9 (82.5)	90.2 (86.5)	90.7 (86.5)	90.3 (84.8)	88.7 (83.3)	89.5 (85.3)
	AD-CLIP (ICCVW23)	✓	✗	✓	84.2 (84.2)	89.4 (86.5)	88.8 (84.5)	89.1 (85.7)	89.5 (86.2)	90.5 (87.3)	88.6 (85.7)
	DAMP (CVPR24)	✓	✗	✓	87.4 (87.4)	89.3 (85.2)	88.8 (85.1)	88.2 (85.1)	88.8 (83.1)	89.1 (84.5)	88.6 (85.1)
	PGA (NeurIPS24)	✓	✗	✓	88.9 (88.9)	84.9 (77.0)	81.8 (71.1)	85.8 (70.9)	74.5 (59.5)	81.3 (75.8)	82.9 (73.9)
	DAPL (TNNLS25)	✓	✗	✓	90.3 (90.3)	89.7 (85.2)	89.7 (88.6)	89.7 (88.8)	89.5 (84.9)	90.2 (86.7)	89.9 (87.4)
	VisTA (Ours)	✓	✓	✓	87.3 (87.3)	90.4 (87.3)	90.6 (87.7)	90.4 (86.1)	91.0 (87.6)	91.5 (87.7)	90.2 (87.3)

Table 11. Classification accuracies (%) on Office-Home with Real World as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
R→A	ProCA (ECCV22)	✓	✓	✗	65.8 (68.8)	72.1 (70.2)	72.9 (69.5)	76.1 (70.4)	75.6 (70.1)	77.4 (69.5)	73.3 (69.8)
	PLDCA (TIP24)	✓	✓	✗	68.9 (74.4)	76.7 (72.6)	78.1 (72.4)	80.1 (71.6)	80.3 (71.8)	80.4 (70.4)	77.4 (72.2)
	CoOp (IJCV22)	–	–	–	79.2 (79.2)	83.1 (79.2)	82.7 (79.2)	83.4 (79.2)	82.5 (79.2)	83.3 (79.2)	82.4 (79.2)
	AttriCLIP (CVPR23)	✗	✓	✓	83.5 (83.5)	85.2 (79.5)	80.5 (75.4)	84.0 (76.7)	81.9 (72.2)	83.9 (78.0)	83.2 (77.5)
	AD-CLIP (ICCVW23)	✓	✗	✓	72.7 (72.7)	85.3 (80.8)	82.4 (79.2)	81.3 (72.4)	82.0 (79.5)	83.3 (74.7)	81.2 (76.6)
	DAMP (CVPR24)	✓	✗	✓	74.7 (74.7)	81.6 (72.7)	79.4 (74.7)	81.2 (74.4)	81.7 (77.5)	80.3 (68.1)	79.8 (73.7)
	PGA (NeurIPS24)	✓	✗	✓	80.5 (80.5)	76.7 (64.6)	37.3 (7.8)	72.2 (60.2)	47.2 (28.6)	59.4 (34.4)	62.2 (46.0)
	DAPL (TNNLS25)	✓	✗	✓	83.5 (83.5)	85.3 (80.0)	81.3 (77.5)	82.8 (77.5)	82.3 (73.9)	82.2 (74.2)	82.9 (77.8)
	VisTA (Ours)	✓	✓	✓	80.3 (80.3)	85.2 (79.2)	84.4 (80.3)	84.8 (80.0)	83.5 (78.5)	84.8 (80.3)	83.8 (79.8)
R→C	ProCA (ECCV22)	✓	✓	✗	50.8 (52.6)	50.6 (49.6)	51.0 (50.8)	52.0 (52.5)	51.9 (50.9)	50.3 (50.9)	51.1 (51.2)
	PLDCA (TIP24)	✓	✓	✗	48.0 (50.2)	60.5 (49.2)	56.5 (48.9)	57.2 (49.6)	56.7 (48.0)	55.8 (46.9)	55.8 (48.8)
	CoOp (IJCV22)	–	–	–	63.1 (63.1)	71.3 (63.1)	71.0 (63.1)	73.0 (63.1)	72.7 (63.1)	71.6 (63.1)	70.4 (63.1)
	AttriCLIP (CVPR23)	✗	✓	✓	66.1 (66.1)	69.1 (56.1)	57.0 (47.7)	62.8 (53.1)	72.2 (63.1)	69.0 (62.2)	66.0 (58.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	62.9 (62.9)	72.9 (65.7)	71.5 (67.5)	81.6 (66.6)	72.3 (62.6)	71.4 (62.3)	72.1 (64.6)
	DAMP (CVPR24)	✓	✗	✓	72.5 (72.5)	74.7 (66.7)	70.0 (66.1)	68.9 (60.7)	70.6 (59.3)	69.0 (59.8)	71.0 (64.2)
	PGA (NeurIPS24)	✓	✗	✓	54.7 (54.7)	64.3 (50.2)	64.6 (52.8)	59.2 (42.3)	61.0 (50.0)	63.1 (51.9)	61.1 (50.3)
	DAPL (TNNLS25)	✓	✗	✓	64.4 (64.4)	72.4 (62.6)	66.8 (59.9)	68.8 (63.7)	71.0 (61.7)	70.4 (62.0)	69.0 (62.4)
	VisTA (Ours)	✓	✓	✓	61.4 (61.4)	71.8 (62.3)	70.3 (62.8)	72.6 (62.2)	73.3 (64.1)	72.2 (64.3)	70.3 (62.8)
R→P	ProCA (ECCV22)	✓	✓	✗	87.4 (90.8)	88.8 (89.0)	88.8 (89.7)	87.9 (88.6)	89.1 (88.0)	87.9 (88.4)	88.3 (89.1)
	PLDCA (TIP24)	✓	✓	✗	83.9 (85.8)	86.9 (84.9)	86.9 (85.3)	88.3 (84.6)	89.4 (84.2)	90.0 (83.6)	87.6 (84.7)
	CoOp (IJCV22)	–	–	–	90.1 (90.1)	89.4 (90.1)	91.6 (90.1)	92.2 (90.1)	92.2 (90.1)	91.9 (90.1)	91.2 (90.1)
	AttriCLIP (CVPR23)	✗	✓	✓	93.7 (93.7)	91.1 (90.6)	92.2 (90.3)	92.4 (90.6)	92.6 (88.9)	90.4 (80.7)	92.1 (89.1)
	AD-CLIP (ICCVW23)	✓	✗	✓	87.0 (87.0)	91.6 (88.9)	93.4 (89.8)	91.9 (88.8)	91.7 (88.2)	92.7 (87.6)	91.4 (88.4)
	DAMP (CVPR24)	✓	✗	✓	91.2 (91.2)	88.8 (89.8)	90.4 (88.2)	91.3 (88.8)	91.0 (89.8)	90.7 (84.0)	90.6 (88.6)
	PGA (NeurIPS24)	✓	✗	✓	89.2 (89.2)	82.7 (75.9)	78.9 (58.6)	86.8 (83.0)	87.6 (79.4)	82.1 (74.1)	84.5 (76.7)
	DAPL (TNNLS25)	✓	✗	✓	90.7 (90.7)	90.7 (87.1)	88.6 (85.9)	89.5 (87.6)	90.2 (84.6)	90.8 (87.1)	90.1 (87.2)
	VisTA (Ours)	✓	✓	✓	91.5 (91.5)	91.6 (91.9)	92.0 (91.6)	92.8 (91.3)	92.8 (90.9)	92.8 (91.5)	92.2 (91.5)

Table 12. Classification accuracies (%) on Mini-DomianNet with Clipart as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
C→P	ProCA (ECCV22)	✓	✓	✗	55.0 (55.0)	54.0 (49.0)	52.0 (44.0)	51.2 (43.0)	49.8 (43.0)	50.7 (42.0)	52.1 (46.0)
	PLDCA (TIP24)	✓	✓	✗	51.3 (56.0)	57.0 (52.1)	60.0 (48.8)	53.2 (43.9)	55.0 (46.1)	56.2 (41.2)	55.4 (48.0)
	CoOp (IJCV22)	-	-	-	74.0 (70.4)	81.0 (70.4)	80.0 (70.4)	78.2 (70.4)	77.0 (70.4)	78.5 (70.4)	78.1 (70.4)
	AttriCLIP (CVPR23)	✗	✓	✓	73.0 (65.2)	78.5 (69.5)	76.7 (66.6)	75.8 (70.3)	76.2 (68.1)	73.8 (65.5)	75.7 (67.5)
	AD-CLIP (ICCVW23)	✓	✗	✓	78.0 (73.9)	82.5 (72.9)	79.0 (68.4)	80.5 (71.1)	79.4 (71.9)	77.5 (69.4)	79.5 (71.3)
	DAMP (CVPR24)	✓	✗	✓	79.0 (71.1)	79.5 (70.2)	79.7 (69.1)	77.8 (69.9)	78.4 (70.9)	77.8 (72.4)	78.7 (70.6)
	PGA (NeurIPS24)	✓	✗	✓	15.6 (15.6)	31.4 (14.5)	48.6 (15.0)	63.7 (16.0)	77.0 (15.0)	80.9 (15.8)	52.9 (15.3)
	DAPL (TNNLS25)	✓	✗	✓	82.0 (76.5)	83.0 (74.9)	81.7 (73.2)	81.2 (74.4)	80.8 (73.3)	81.5 (72.5)	81.7 (74.1)
	VisTA (Ours)	✓	✓	✓	81.0 (75.7)	85.5 (76.2)	85.3 (76.2)	84.5 (76.0)	84.0 (76.0)	84.0 (76.3)	84.0 (76.1)
C→R	ProCA (ECCV22)	✓	✓	✗	76.0 (76.0)	77.5 (73.0)	77.3 (70.0)	73.8 (72.0)	73.0 (68.0)	73.3 (66.0)	75.2 (70.8)
	PLDCA (TIP24)	✓	✓	✗	62.5 (65.0)	68.5 (64.0)	71.7 (61.2)	71.0 (59.1)	69.0 (55.5)	71.5 (54.8)	69.0 (59.9)
	CoOp (IJCV22)	-	-	-	88.0 (88.1)	88.5 (88.1)	90.0 (88.1)	87.8 (88.1)	87.6 (88.1)	88.5 (88.1)	88.4 (88.1)
	AttriCLIP (CVPR23)	✗	✓	✓	68.0 (65.2)	80.0 (84.6)	81.0 (81.6)	79.8 (78.9)	77.2 (80.8)	74.7 (70.5)	76.8 (76.9)
	AD-CLIP (ICCVW23)	✓	✗	✓	88.0 (88.0)	86.2 (88.0)	89.2 (90.0)	88.8 (89.5)	90.0 (89.0)	89.5 (88.0)	88.6 (88.8)
	DAMP (CVPR24)	✓	✗	✓	87.0 (85.3)	84.5 (88.8)	85.0 (86.1)	87.2 (84.7)	86.4 (84.2)	85.0 (80.9)	85.9 (85.0)
	PGA (NeurIPS24)	✓	✗	✓	18.8 (18.8)	35.7 (18.0)	54.1 (18.0)	70.7 (17.6)	89.3 (18.2)	90.2 (16.8)	59.8 (17.9)
	DAPL (TNNLS25)	✓	✗	✓	98.0 (93.1)	92.0 (92.4)	92.7 (92.1)	91.8 (92.1)	91.2 (91.6)	90.3 (91.8)	92.7 (92.2)
	VisTA (Ours)	✓	✓	✓	93.0 (92.1)	92.0 (92.1)	93.0 (92.2)	91.2 (92.0)	91.4 (92.0)	91.5 (92.2)	92.0 (92.1)
C→S	ProCA (ECCV22)	✓	✓	✗	61.0 (61.0)	55.0 (58.0)	51.0 (54.0)	50.2 (56.0)	51.6 (55.0)	51.0 (52.0)	53.3 (56.0)
	PLDCA (TIP24)	✓	✓	✗	57.2 (53.6)	57.7 (52.5)	57.9 (43.3)	52.2 (42.9)	50.3 (38.9)	51.9 (38.5)	54.5 (44.9)
	CoOp (IJCV22)	-	-	-	78.0 (76.2)	77.0 (76.2)	79.7 (76.2)	79.2 (76.2)	77.4 (76.2)	77.8 (76.2)	78.2 (76.2)
	AttriCLIP (CVPR23)	✗	✓	✓	72.0 (74.0)	72.5 (76.1)	75.0 (76.8)	74.8 (77.0)	76.4 (77.4)	76.7 (75.9)	74.6 (76.2)
	AD-CLIP (ICCVW23)	✓	✗	✓	79.0 (78.9)	75.0 (77.9)	82.0 (80.2)	76.5 (75.8)	76.0 (72.9)	75.7 (75.4)	77.4 (76.9)
	DAMP (CVPR24)	✓	✗	✓	78.0 (81.5)	79.5 (78.5)	80.7 (76.9)	78.5 (77.2)	76.6 (76.1)	76.5 (80.1)	78.3 (78.4)
	PGA (NeurIPS24)	✓	✗	✓	15.6 (15.6)	29.7 (14.5)	48.0 (14.6)	59.6 (13.7)	74.8 (14.5)	80.3 (14.6)	51.3 (14.6)
	DAPL (TNNLS25)	✓	✗	✓	77.0 (81.6)	79.5 (80.9)	82.3 (80.6)	80.8 (78.9)	79.0 (78.7)	78.3 (78.6)	79.5 (79.9)
	VisTA (Ours)	✓	✓	✓	80.0 (81.4)	79.0 (81.2)	83.0 (81.1)	83.0 (81.0)	82.2 (80.4)	81.5 (80.3)	81.5 (80.9)

Table 13. Classification accuracies (%) on Mini-DomianNet with Painting as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
P→C	ProCA (ECCV22)	✓	✓	✗	63.0 (63.0)	55.5 (56.0)	62.0 (58.0)	63.2 (59.0)	57.0 (58.0)	56.2 (56.0)	59.5 (58.3)
	PLDCA (TIP24)	✓	✓	✗	65.0 (65.3)	63.6 (59.6)	66.3 (56.6)	65.4 (56.7)	66.0 (56.3)	61.5 (55.5)	64.6 (58.3)
	CoOp (IJCV22)	-	-	-	81.0 (80.9)	83.5 (80.9)	87.0 (80.9)	87.0 (80.9)	84.6 (80.9)	82.5 (80.9)	84.3 (80.9)
	AttriCLIP (CVPR23)	✗	✓	✓	72.0 (71.8)	79.0 (79.4)	84.3 (80.4)	87.2 (82.1)	85.4 (80.1)	81.8 (80.5)	81.6 (79.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	83.4 (81.0)	84.0 (80.5)	86.7 (81.6)	90.0 (80.3)	84.8 (80.3)	80.6 (85.7)	84.9 (81.6)
	DAMP (CVPR24)	✓	✗	✓	83.0 (82.5)	89.0 (82.4)	89.7 (82.4)	90.0 (81.2)	85.4 (82.2)	82.7 (81.9)	86.6 (82.1)
	PGA (NeurIPS24)	✓	✗	✓	16.6 (16.6)	34.2 (16.0)	50.2 (15.2)	43.4 (6.4)	82.6 (14.5)	80.7 (14.1)	51.3 (13.8)
	DAPL (TNNLS25)	✓	✗	✓	77.0 (83.4)	86.5 (83.2)	87.3 (82.8)	88.2 (82.4)	86.4 (82.0)	85.5 (81.7)	85.1 (82.6)
	VisTA (Ours)	✓	✓	✓	82.0 (83.1)	85.5 (84.0)	88.7 (83.9)	89.5 (83.9)	88.2 (84.1)	85.7 (84.1)	86.6 (83.9)
P→R	ProCA (ECCV22)	✓	✓	✗	88.0 (88.0)	85.5 (84.0)	88.0 (84.0)	85.2 (84.0)	79.8 (75.0)	83.0 (77.0)	84.9 (82.0)
	PLDCA (TIP24)	✓	✓	✗	73.9 (76.2)	75.7 (75.5)	78.3 (74.2)	78.6 (70.7)	78.2 (69.8)	78.6 (67.8)	77.2 (72.4)
	CoOp (IJCV22)	-	-	-	90.0 (89.9)	87.0 (89.9)	89.3 (89.9)	88.5 (89.9)	88.6 (89.9)	88.8 (89.9)	88.7 (89.9)
	AttriCLIP (CVPR23)	✗	✓	✓	75.0 (70.9)	82.0 (87.2)	86.0 (87.9)	84.2 (85.4)	83.6 (84.1)	84.0 (84.1)	82.5 (83.3)
	AD-CLIP (ICCVW23)	✓	✗	✓	92.1 (94.0)	91.1 (89.0)	89.6 (91.7)	90.8 (91.4)	89.2 (90.3)	90.8 (91.2)	90.6 (91.3)
	DAMP (CVPR24)	✓	✗	✓	88.0 (85.2)	91.0 (90.4)	89.3 (89.0)	87.0 (87.5)	87.0 (87.7)	87.3 (86.1)	88.3 (87.7)
	PGA (NeurIPS24)	✓	✗	✓	18.6 (18.6)	35.9 (18.2)	54.3 (17.6)	70.9 (18.2)	87.5 (18.0)	89.3 (17.6)	59.4 (18.0)
	DAPL (TNNLS25)	✓	✗	✓	96.0 (93.1)	92.5 (92.5)	92.7 (91.9)	92.0 (92.0)	91.4 (91.9)	91.0 (91.2)	92.6 (92.1)
	VisTA (Ours)	✓	✓	✓	95.0 (92.6)	93.0 (92.9)	93.3 (93.1)	91.8 (93.1)	91.2 (93.3)	91.7 (93.3)	92.7 (93.1)
P→S	ProCA (ECCV22)	✓	✓	✗	62.0 (62.0)	31.0 (45.0)	25.3 (45.0)	40.2 (52.0)	37.6 (50.0)	37.0 (42.0)	38.9 (49.3)
	PLDCA (TIP24)	✓	✓	✗	64.4 (56.8)	60.0 (52.1)	60.0 (45.3)	55.4 (43.5)	53.1 (44.1)	52.5 (43.1)	57.6 (47.5)
	CoOp (IJCV22)	-	-	-	79.0 (75.0)	76.0 (75.0)	75.7 (75.0)	74.2 (75.0)	74.6 (75.0)	73.3 (75.0)	75.5 (75.0)
	AttriCLIP (CVPR23)	✗	✓	✓	77.0 (78.1)	75.5 (79.0)	81.0 (79.8)	79.5 (80.4)	80.0 (80.4)	78.2 (78.7)	78.5 (79.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	77.0 (79.4)	75.5 (79.1)	79.3 (78.7)	77.4 (78.8)	80.8 (80.1)	77.5 (77.8)	77.9 (79.0)
	DAMP (CVPR24)	✓	✗	✓	83.0 (83.2)	80.5 (80.0)	81.7 (80.6)	80.8 (79.2)	80.8 (81.2)	79.2 (79.8)	81.0 (80.7)
	PGA (NeurIPS24)	✓	✗	✓	15.6 (15.6)	23.6 (9.2)	45.5 (15.0)	59.0 (11.9)	75.4 (14.6)	76.2 (14.8)	49.2 (13.5)
	DAPL (TNNLS25)	✓	✗	✓	75.0 (81.4)	80.0 (80.7)	81.0 (79.4)	80.0 (78.1)	78.6 (79.8)	78.0 (79.7)	78.8 (79.9)
	VisTA (Ours)	✓	✓	✓	80.0 (82.3)	80.0 (82.2)	83.3 (81.9)	83.2 (82.0)	82.2 (81.8)	81.7 (81.8)	81.7 (82.0)

Table 14. Classification accuracies (%) on Mini-DomianNet with Real World as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
R→C	ProCA (ECCV22)	✓	✓	✗	57.0 (57.0)	59.5 (58.0)	59.7 (53.0)	58.0 (56.0)	58.6 (54.0)	56.8 (47.0)	58.3 (54.2)
	PLDCA (TIP24)	✓	✓	✗	59.6 (57.7)	67.2 (54.4)	66.8 (52.9)	64.3 (49.9)	63.7 (50.8)	60.5 (48.8)	63.7 (52.4)
	CoOp (IJCV22)	-	-	-	75.0 (81.1)	84.5 (81.1)	87.3 (81.1)	88.0 (81.1)	85.4 (81.1)	83.0 (81.1)	83.9 (81.1)
	AttriCLIP (CVPR23)	✗	✓	✓	79.0 (73.7)	82.0 (80.2)	86.7 (80.9)	88.2 (80.8)	84.4 (80.7)	83.0 (79.8)	83.9 (79.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	80.0 (82.5)	88.0 (82.7)	87.0 (81.1)	88.0 (82.3)	86.8 (81.3)	80.7 (83.8)	85.1 (82.3)
	DAMP (CVPR24)	✓	✗	✓	81.0 (82.4)	83.5 (79.3)	89.3 (82.2)	87.5 (78.7)	84.0 (74.7)	79.7 (76.8)	84.2 (79.2)
	PGA (NeurIPS24)	✓	✗	✓	15.0 (15.0)	24.2 (8.6)	44.9 (16.0)	64.3 (13.3)	78.7 (13.1)	65.8 (14.1)	48.8 (13.3)
	DAPL (TNNLS25)	✓	✗	✓	81.0 (82.9)	87.0 (83.4)	88.7 (82.3)	88.8 (82.0)	85.6 (81.1)	85.0 (80.8)	86.0 (82.1)
	VisTA (Ours)	✓	✓	✓	81.0 (83.3)	86.5 (83.6)	88.3 (83.3)	89.0 (83.6)	87.8 (83.4)	85.0 (83.3)	86.3 (83.4)
R→P	ProCA (ECCV22)	✓	✓	✗	51.0 (51.0)	51.5 (48.0)	59.3 (48.0)	55.8 (46.0)	50.2 (42.0)	55.2 (47.0)	53.8 (47.0)
	PLDCA (TIP24)	✓	✓	✗	57.9 (57.0)	63.5 (56.2)	67.7 (53.0)	63.7 (51.0)	63.7 (49.9)	66.2 (51.3)	63.8 (53.1)
	CoOp (IJCV22)	-	-	-	72.0 (71.1)	79.0 (71.1)	79.7 (71.1)	78.8 (71.1)	78.2 (71.1)	77.8 (71.1)	77.6 (71.1)
	AttriCLIP (CVPR23)	✗	✓	✓	74.0 (64.0)	75.5 (66.2)	72.3 (68.2)	73.8 (72.0)	77.8 (70.9)	70.5 (65.2)	74.0 (67.7)
	AD-CLIP (ICCVW23)	✓	✗	✓	72.0 (69.9)	82.0 (72.0)	80.7 (68.2)	80.2 (70.4)	77.2 (71.2)	78.7 (69.5)	78.5 (70.2)
	DAMP (CVPR24)	✓	✗	✓	77.0 (69.0)	81.5 (72.7)	81.0 (71.2)	78.0 (71.7)	78.8 (71.1)	74.0 (71.9)	78.4 (71.3)
	PGA (NeurIPS24)	✓	✗	✓	14.1 (14.1)	27.3 (10.5)	46.1 (13.7)	61.7 (14.8)	64.6 (15.0)	75.6 (14.5)	48.2 (13.8)
	DAPL (TNNLS25)	✓	✗	✓	81.0 (76.5)	83.0 (74.7)	82.7 (73.8)	80.8 (74.1)	81.0 (73.3)	81.0 (73.1)	81.6 (74.2)
	VisTA (Ours)	✓	✓	✓	79.0 (75.5)	83.0 (75.9)	83.3 (76.4)	82.2 (76.4)	83.4 (76.2)	83.0 (76.3)	82.3 (76.1)
R→S	ProCA (ECCV22)	✓	✓	✗	37.0 (37.0)	38.5 (34.0)	41.3 (37.0)	44.0 (40.0)	41.6 (36.0)	39.7 (35.0)	40.4 (36.5)
	PLDCA (TIP24)	✓	✓	✗	47.8 (39.6)	46.5 (34.8)	50.0 (31.5)	48.4 (31.0)	42.3 (28.5)	42.9 (27.6)	46.3 (32.2)
	CoOp (IJCV22)	-	-	-	76.0 (76.7)	78.0 (76.7)	81.0 (76.7)	82.0 (76.7)	81.4 (76.7)	79.2 (76.7)	79.6 (76.7)
	AttriCLIP (CVPR23)	✗	✓	✓	71.0 (73.4)	76.5 (77.2)	78.7 (78.0)	77.0 (77.8)	78.6 (76.2)	73.3 (75.6)	75.8 (76.4)
	AD-CLIP (ICCVW23)	✓	✗	✓	81.0 (79.1)	80.5 (80.6)	81.7 (78.2)	80.0 (77.5)	82.4 (79.2)	79.8 (80.3)	80.9 (79.2)
	DAMP (CVPR24)	✓	✗	✓	78.0 (83.1)	81.0 (79.6)	80.3 (77.9)	82.0 (76.0)	77.0 (74.7)	78.5 (78.5)	79.5 (78.3)
	PGA (NeurIPS24)	✓	✗	✓	14.8 (14.8)	26.2 (11.5)	40.0 (10.9)	58.0 (14.6)	66.2 (8.2)	76.0 (13.7)	46.9 (12.3)
	DAPL (TNNLS25)	✓	✗	✓	77.0 (81.4)	78.0 (80.0)	82.0 (79.6)	80.0 (78.9)	79.2 (79.7)	77.3 (78.9)	78.9 (79.8)
	VisTA (Ours)	✓	✓	✓	79.0 (81.7)	81.0 (81.5)	83.7 (81.9)	82.5 (81.5)	82.2 (81.4)	81.2 (80.8)	81.6 (81.5)

Table 15. Classification accuracies (%) on Mini-DomianNet with Sketch as source domain. Note that the results outside the brackets are Step-level Accuracy, while the results in brackets represent the S-1 Accuracy in each time step. DA, CI, and RF respectively represent domain adaptation, class-incremental, and rehearsal-free.

Task	Method	DA	CI	RF	Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.
S→C	ProCA (ECCV22)	✓	✓	✗	63.0 (63.0)	57.0 (57.0)	61.7 (55.0)	59.8 (53.0)	56.2 (51.0)	57.7 (53.0)	59.2 (55.3)
	PLDCA (TIP24)	✓	✓	✗	65.5 (68.9)	53.1 (62.5)	63.5 (61.4)	60.5 (59.1)	57.0 (54.1)	62.1 (57.1)	60.3 (60.5)
	CoOp (IJCV22)	-	-	-	85.0 (82.8)	80.5 (82.8)	84.3 (82.8)	85.5 (82.8)	84.0 (82.8)	82.8 (82.8)	83.7 (82.8)
	AttriCLIP (CVPR23)	✗	✓	✓	78.0 (78.9)	80.5 (82.0)	87.0 (82.5)	87.0 (82.6)	86.6 (83.5)	83.8 (82.5)	83.8 (82.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	75.0 (77.1)	84.5 (82.0)	87.0 (81.5)	86.2 (81.2)	86.8 (84.2)	85.2 (83.1)	84.1 (81.5)
	DAMP (CVPR24)	✓	✗	✓	86.0 (84.0)	87.0 (83.3)	88.3 (83.3)	87.8 (79.8)	86.2 (79.3)	84.3 (82.6)	86.6 (82.1)
	PGA (NeurIPS24)	✓	✗	✓	14.1 (14.1)	27.9 (11.3)	52.1 (16.6)	44.9 (10.5)	79.5 (10.5)	84.4 (15.2)	50.5 (13.0)
	DAPL (TNNLS25)	✓	✗	✓	81.0 (83.6)	88.0 (83.8)	89.0 (82.8)	89.5 (81.9)	86.4 (80.6)	85.7 (81.0)	86.6 (82.3)
	VisTA (Ours)	✓	✓	✓	78.0 (83.1)	85.5 (83.9)	88.0 (84.3)	88.8 (84.0)	87.8 (83.8)	86.0 (83.7)	85.7 (83.8)
S→P	ProCA (ECCV22)	✓	✓	✗	28.0 (28.0)	28.5 (26.0)	43.3 (34.0)	34.5 (27.0)	38.0 (33.0)	44.3 (31.0)	36.1 (29.8)
	PLDCA (TIP24)	✓	✓	✗	53.3 (50.5)	51.0 (49.5)	52.2 (41.7)	54.4 (42.2)	56.6 (42.7)	58.8 (39.3)	54.4 (44.3)
	CoOp (IJCV22)	-	-	-	77.0 (71.8)	76.0 (71.8)	77.3 (71.8)	76.0 (71.8)	76.0 (71.8)	76.5 (71.8)	76.5 (71.8)
	AttriCLIP (CVPR23)	✗	✓	✓	76.0 (66.2)	74.0 (68.4)	75.7 (70.3)	76.0 (70.1)	77.2 (71.0)	78.0 (71.8)	76.1 (69.6)
	AD-CLIP (ICCVW23)	✓	✗	✓	78.0 (71.0)	80.0 (70.0)	79.7 (70.6)	78.5 (71.9)	80.2 (73.0)	79.5 (72.5)	79.3 (71.5)
	DAMP (CVPR24)	✓	✗	✓	79.0 (72.5)	79.0 (73.8)	78.7 (73.7)	76.2 (70.8)	78.0 (71.2)	74.7 (72.0)	77.6 (72.3)
	PGA (NeurIPS24)	✓	✗	✓	14.5 (14.5)	32.6 (15.0)	47.3 (15.8)	61.7 (15.6)	71.9 (14.3)	81.8 (14.8)	51.6 (15.0)
	DAPL (TNNLS25)	✓	✗	✓	82.0 (76.9)	84.5 (74.4)	81.3 (73.4)	81.0 (74.4)	80.2 (73.6)	80.2 (72.5)	81.5 (74.2)
	VisTA (Ours)	✓	✓	✓	80.0 (75.4)	83.0 (75.9)	83.0 (76.2)	82.0 (76.3)	81.6 (76.3)	82.3 (76.4)	82.0 (76.1)
S→R	ProCA (ECCV22)	✓	✓	✗	77.0 (77.0)	70.5 (73.0)	72.3 (70.0)	69.2 (67.0)	68.2 (65.0)	70.8 (64.0)	71.3 (69.3)
	PLDCA (TIP24)	✓	✓	✗	73.4 (75.6)	67.5 (71.2)	70.2 (69.2)	69.2 (64.4)	69.4 (61.7)	71.9 (61.5)	70.3 (67.3)
	CoOp (IJCV22)	-	-	-	88.0 (88.1)	83.0 (88.1)	87.0 (88.1)	87.0 (88.1)	86.0 (88.1)	86.2 (88.1)	86.2 (88.1)
	AttriCLIP (CVPR23)	✗	✓	✓	74.0 (68.3)	79.0 (84.4)	78.0 (81.7)	75.5 (77.2)	77.0 (77.3)	78.2 (78.9)	77.0 (78.0)
	AD-CLIP (ICCVW23)	✓	✗	✓	95.0 (92.2)	89.0 (89.0)	90.7 (90.8)	90.0 (89.8)	90.0 (90.8)	89.8 (90.1)	90.8 (90.5)
	DAMP (CVPR24)	✓	✗	✓	91.0 (90.7)	85.5 (88.5)	86.3 (86.3)	86.0 (83.6)	88.4 (86.9)	82.3 (81.4)	86.6 (86.2)
	PGA (NeurIPS24)	✓	✗	✓	18.2 (18.2)	35.0 (17.6)	53.3 (17.4)	71.3 (18.2)	87.7 (17.8)	89.6 (18.2)	59.2 (17.9)
	DAPL (TNNLS25)	✓	✗	✓	97.0 (93.2)	93.0 (92.4)	94.7 (92.4)	91.8 (92.2)	91.8 (92.0)	90.8 (91.9)	93.2 (92.4)
	VisTA (Ours)	✓	✓	✓	94.0 (92.1)	91.5 (92.3)	93.3 (92.7)	91.8 (92.7)	91.6 (92.8)	91.2 (92.9)	92.2 (92.6)

Table 16. Final Accuracy (%) of CI-SFUDA on Office-31. DA, CI, Sf, and RF respectively represent domain adaptation, class-incremental, source-free, and rehearsal-free.

Method	DA	CI	Sf	RF	A→D	A→W	D→A	D→W	W→A	W→D	Avg.
GROTO (CVPR25)	✓	✓	✓	✗	99.4	99.0	81.3	99.0	81.3	98.1	93.0
VisTA (Ours)	✓	✓	✓	✓	87.4	88.8	85.2	93.5	82.5	95.7	88.9

Table 17. Final Accuracy (%) of CI-SFUDA on Office-Home. DA, CI, Sf, and RF respectively represent domain adaptation, class-incremental, source-free, and rehearsal-free.

Method	DA	CI	Sf	RF	A→C	A→P	A→R	C→A	C→P	C→R	P→A	P→C	P→R	R→A	R→C	R→P	Avg.
GROTO (CVPR25)	✓	✓	✓	✗	65.7	86.4	89.7	85.8	86.3	90.9	86.0	67.1	90.1	86.9	66.2	89.3	82.5
VisTA (Ours)	✓	✓	✓	✓	71.4	92.3	90.8	84.5	92.2	90.5	83.7	72.5	91.1	83.5	71.8	92.6	84.7

Table 18. Final Accuracy (%) of CI-SFUDA on Mini-DomainNet. DA, CI, Sf, and RF respectively represent domain adaptation, class-incremental, source-free, and rehearsal-free.

Method	DA	CI	Sf	RF	P→C	R→C	S→C	C→P	R→P	S→P	C→R	P→R	S→R	C→S	P→S	R→S	Avg.
GROTO (CVPR25)	✓	✓	✓	✗		66.2		61.7			78.0				56.7		65.7
VisTA (Ours)	✓	✓	✓	✓	85.7	85.5	86.2	82.7	81.0	81.3	91.5	90.8	91.2	81.7	78.8	80.7	84.8

Table 19. Step-level Accuracy (%) of CI-SFUDA on Office-31 and Office-Home. DA, CI, Sf, and RF respectively represent domain adaptation, class-incremental, source-free, and rehearsal-free.

Method	DA	CI	Sf	RF	Office-31					Office-Home							
					Step 1	Step 2	Step 3	Avg.		Step 1	Step 2	Step 3	Step 4	Step 5	Step 6	Avg.	
GROTO (CVPR25)	✓	✓	✓	✗	96.7	94.9	93.0	94.9		84.4	84.8	81.7	83.3	82.8	82.5	83.2	
VisTA (Ours)	✓	✓	✓	✓	93.9	92.7	88.9	91.8		81.0	84.3	83.6	84.6	84.7	84.7	83.8	

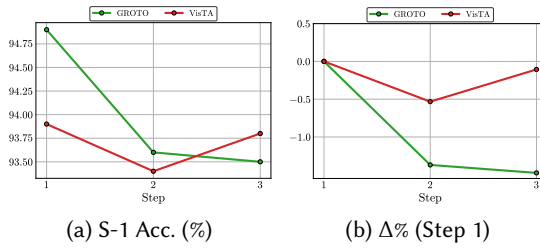


Fig. 3. S-1 Accuracy of CI-SFUDA at each step and its percentage change ($\Delta\%$) compared with Step-1 on Office-31.

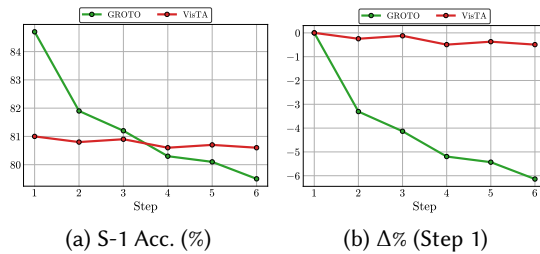


Fig. 4. S-1 Accuracy of CI-SFUDA at each step and its percentage change ($\Delta\%$) compared with Step-1 on Office-Home.

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